

EXERCISE - 7

Transfer Files from One Virtual Machine to Another

Aim:

To create two virtual machines using **VirtualBox** and **Vagrant**, establish network connectivity between them, and transfer a file securely from one virtual machine to another.

Algorithm:

1. Install and configure VirtualBox and Vagrant on the host system.
2. Create a virtual network for communication between the virtual machines.
3. Create a Vagrant project directory and initialize it using a Vagrantfile.
4. Configure two Ubuntu virtual machines named **vm1** and **vm2**.
5. Assign the IP addresses **192.168.56.10** to VM1 and **192.168.56.11** to VM2.
6. Start both virtual machines using Vagrant.
7. Connect to each virtual machine using SSH and verify their assigned IP addresses.
8. Test network connectivity by pinging VM1 from VM2.
9. Create a sample text file named **transfer.txt** in VM1.
10. Use **SCP (Secure Copy Protocol)** to securely transfer **transfer.txt** from VM1 to VM2.
11. Log in to VM2 and verify that the transferred file is present.
12. Display the contents of the transferred file to confirm successful file transfer.
13. Collect the completed Cloudlets and display their status, VM ID, execution time, start time, and finish time.
14. Verify that the shortest Cloudlet completes first, followed by the medium and longest Cloudlets.

15. Test each menu option and exit the application successfully.

Output:

Network connectivity was successfully established:

```
[vagrant@vm2:~]$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a3:68:45 brd ff:ff:ff:ff:ff:ff
    altname enp0s8
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic eth0
        valid_lft 86159sec preferred_lft 86159sec
    inet6 fd17:625c:f037:2:a00:27ff:fea3:6845/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86161sec preferred_lft 14161sec
    inet6 fe80::a00:27ff:fea3:6845/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:64:7d:bd brd ff:ff:ff:ff:ff:ff
    altname enp0s9
    inet 192.168.56.11/24 brd 192.168.56.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fd47:2488:fc2a:5557:a00:27ff:fe64:7dbd/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591944sec preferred_lft 604744sec
    inet6 fe80::a00:27ff:fe64:7dbd/64 scope link
        valid_lft forever preferred_lft forever
[vagrant@vm2:~]$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=4.62 ms
```

The file was transferred using SCP:

```
Last login: Tue Sep 15 10:10:00 2020 from 10.0.2.2
[vagrant@vm1:~]$ ls
transfer.txt
[vagrant@vm1:~]$ scp transfer.txt vagrant@192.168.56.11:/home/vagrant/
[vagrant@192.168.56.11's password:
transfer.txt                                     100%  39  184.1KB/s   00:00
[vagrant@vm1:~]$ exit
logout
[lokaa@Lokaas-MacBook-Air IaaS_Lab1 % vagrant ssh vm2
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic aarch64)
```

The transferred file was verified in VM2:

```
[vagrant@vm2:~$ ls
transfer.txt
[vagrant@vm2:~$ cat transfer.txt
Hello from VM1 – Cloud Computing Lab 7
```

Result:

Thus, two Ubuntu virtual machines were successfully created and connected through a virtual network. Network communication between the VMs was verified, and a file was **successfully transferred from VM1 to VM2 using SCP**.